

Label coverage in the smart-meter data

What the asset register can and cannot tell us

From 1,192 register entries to 208 usable meters

Each stage removes labels for a different, identifiable reason

878

subsidised customers

1,192 register rows, but 289 are repeat installations by the same customer

337

linked to a meter

541 customers have no meter listed in the delivered mapping

208

meters we can train on

after removing short histories and ambiguous multi-meter customers

The middle number is the one worth acting on. The other two losses are inherent to the data; that one looks recoverable.

Stage 1 — the register lists installations, not households

GIGI register → distinct customers

1192

rows in the asset register

1167

25 rows carry no partner number

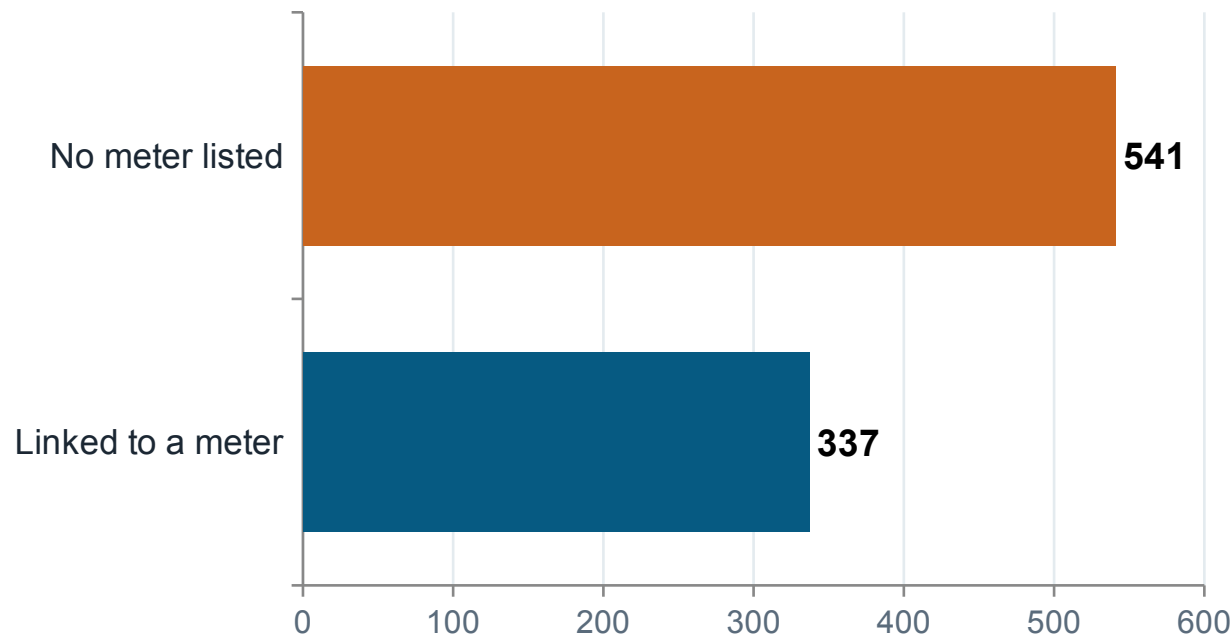
878

distinct Geschäftspartner

- 289 rows are repeat installations by the same customer
- Typical pattern: PV subsidised in 2019, a battery added in 2023
- This is not lost information — it gives each asset its own date

Stage 2 — 62% of subsidised customers have no meter link

This single gap costs more labels than every other stage combined



Not a formatting problem

- Unmatched partner numbers use the same 6-digit format
- They fall in the same numeric range (10179–920285)
- The mapping file covers 76,718 partners — these 541 are simply absent

Likely cause: assets are subsidised across a wider area than the grid is operated for, or the extract was limited to smart-metered customers.

Stage 3 — meeting the measurements

Linked meters → meters with enough clean history

413

Zählpunkte behind those 337 customers

411

present in the load-profile data

281

with at least one year of readings

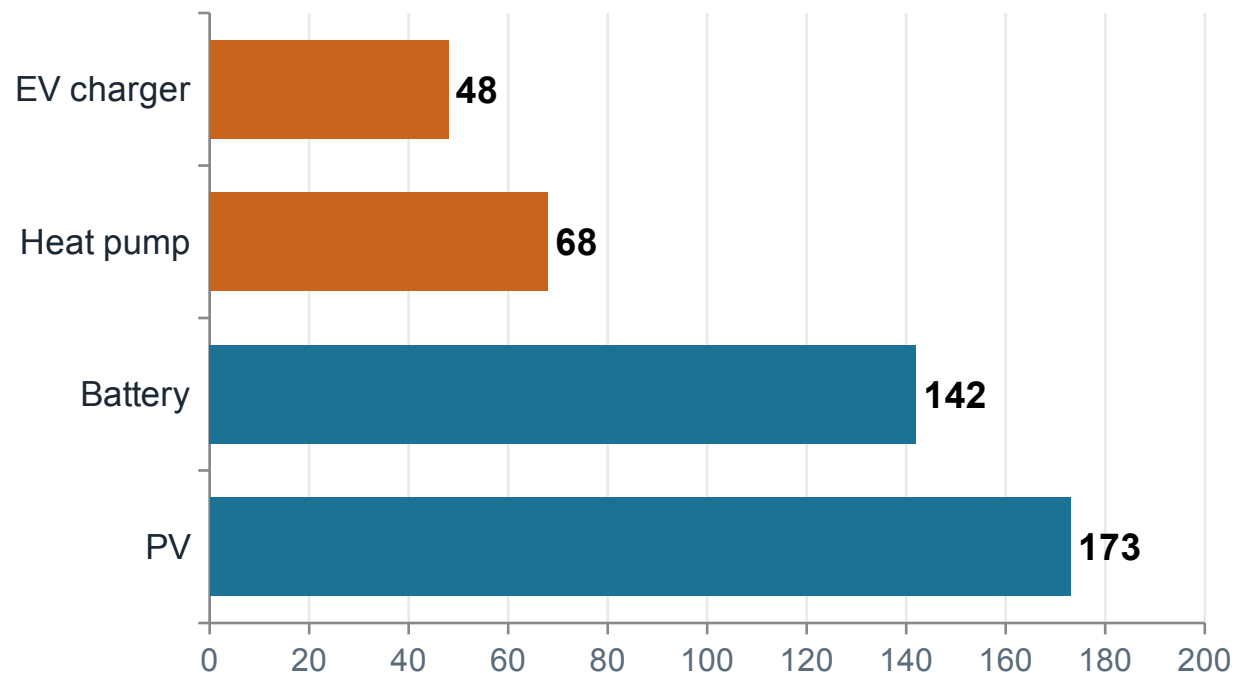
208

customer owns exactly one meter

- 130 meters lost to the rollout — 45 first appear in March 2026, 196 cover under half the window
- 73 lost to ambiguity — the register says the household has a heat pump, not which of its meters

What is left to train on, per asset

Positive examples among the 208 usable meters



Enough to measure, not to fit

- With 68 heat pumps and 48 chargers, a model fitted on these would memorise them
- We build detectors from the physics instead — sustained 3.7 / 7.4 / 11 kW charging blocks, thermostat cycling
- The labelled meters then score the detector, which still runs on all 82,476 meters

Five things we found in the data

Each verified against the delivered files

1 Register negatives are not negatives

A dash means "no subsidy record", not "no asset". Of 19 meters marked as having no PV, 13 were exporting.

2 Commissioning dates are approximate

Of 22 PV meters testable against measurements, 11 were already exporting before the recorded date — by 31 to 457 days.

3 A week of data is missing

No readings at all for any meter from 24 to 30 April 2023.

4 Daylight saving is lossy at source

The export writes a fixed 96-slot grid, so the extra October hour is absent and the March day is padded.

5 A duplicated meter-month

March 2026 contains 62 rows that duplicate an existing meter, OBIS code and date.

What would help most

- 1 The missing partner-to-meter links**

Mapping the 541 unlinked customers would multiply the labelled set by 2.6 — the largest single gain available, and it needs no modelling work.
- 2 Confirmation of what the mapping covers**

Is it limited to the area where the grid is operated, or to smart-metered customers? That tells us whether the gap is closable at all.
- 3 Nothing else is blocking**

For PV we date commissioning from measured export rather than the register, which is accurate to within days and needs no labels.